S24-A520U20-1

GCE A LEVEL

A520U20-1

PREDICTED PAPER – HARD

ECONOMICS – A level component 2

Exploring Economic Behaviour

2 hours 30 minutes

A520U201
01

ADDITIONAL MATERIALS

A calculator.

A WJEC pink 16-page answer booklet.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Answer **all** questions.

Write your answers in the separate answer booklet provided, following the instructions on the front of the answer booklet.

Use both sides of the paper. Write only within the white areas of the booklet.

Write the question number in the two boxes in the left-hand margin at the start of each answer, for example **1 1**.

Leave at least two line spaces between each answer.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

You are reminded of the necessity for good English and orderly presentation in your answers.

Answer **all** questions.

Things can only get better?

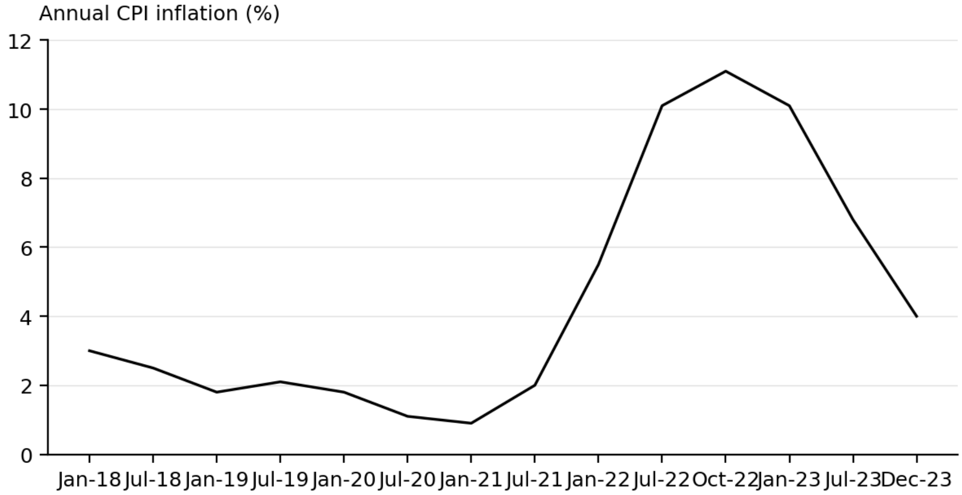
Between 2021 and 2023 the UK economy experienced its sharpest inflationary episode in more than forty years. Headline CPI inflation rose from under 1% in early 2021 to a peak of 11.1% in October 2022 – the highest reading since 1981. The policy response from the Bank of England and HM Treasury tested the theoretical frameworks that have dominated UK macroeconomic policy since 1997.

The inflation shock

The inflationary surge had multiple drivers. Global demand rebounded rapidly following the easing of Covid-19 restrictions; supply chains were dislocated; energy markets were disrupted by Russia's invasion of Ukraine in February 2022; and UK labour supply was constrained by a rise in economic inactivity. Together these forces pushed inflation far above the Bank of England's 2% target (**Figure 1**).

Figure 1

UK CPI inflation, Jan 2018 – Dec 2023

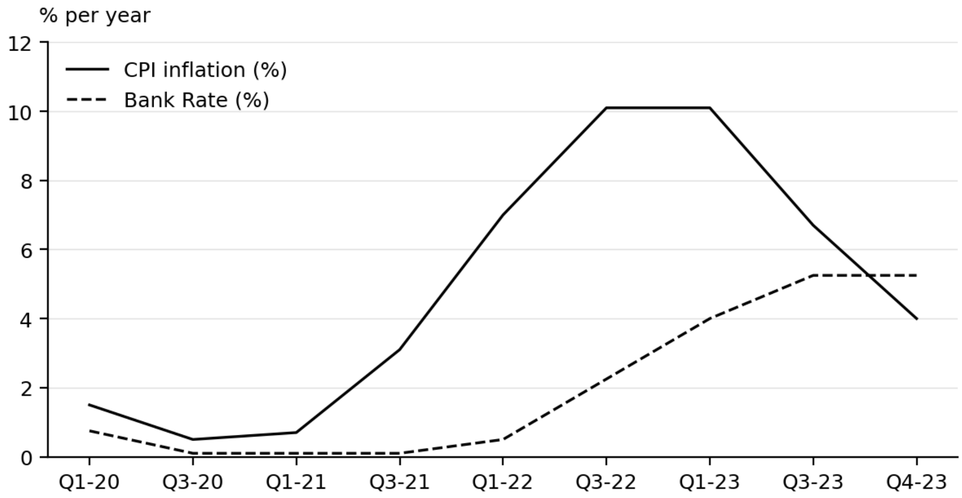


Monetary policy tightening

The Monetary Policy Committee (MPC) kept Bank Rate at its historic low of 0.1% until December 2021. Once it began raising the rate, it did so repeatedly and aggressively, taking Bank Rate from 0.1% in late 2021 to 5.25% by August 2023 – the highest rate since 2008 (**Figure 2**).

Figure 2

UK Bank Rate and CPI inflation, 2020-2023

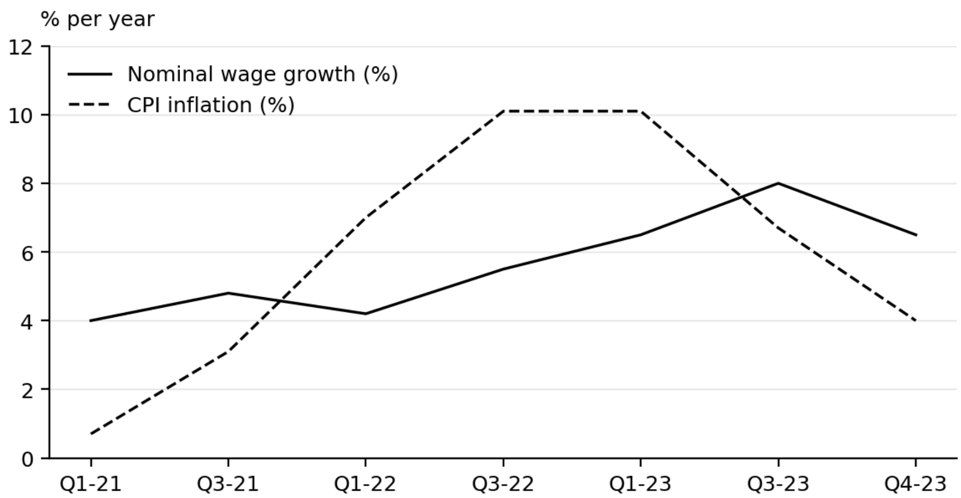


The cost of living and wage response

With inflation far above nominal wage growth for most of the 2022-23 period, real wages fell sharply. Only in late 2023 did nominal wage growth catch up with price rises (**Figure 3**).

Figure 3

Nominal wage growth vs CPI inflation, 2021-2023

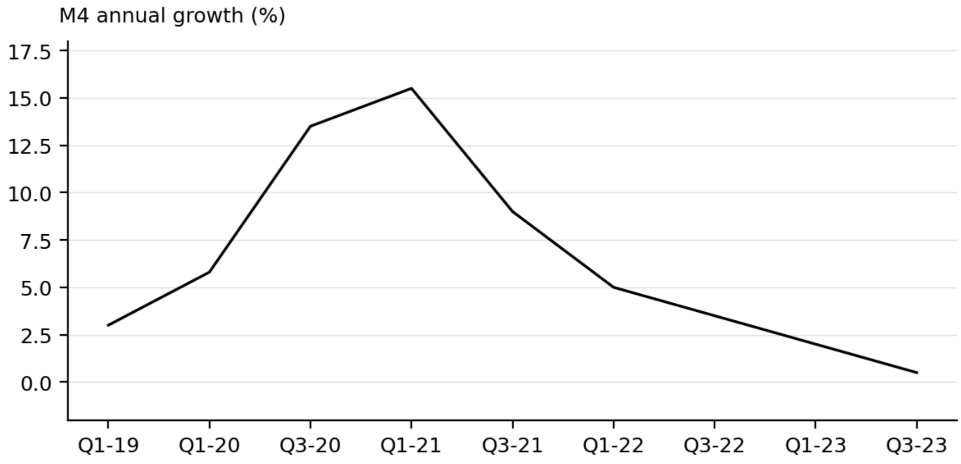


The monetary aggregates

Critics of the Bank of England argue that the expansion of the broad money supply during the pandemic – partly a result of large-scale asset purchases (quantitative easing) – created the conditions for the subsequent inflation (**Figure 4**).

Figure 4

UK broad money (M4) annual growth, 2019-2023



Household confidence and output

The cost-of-living shock weighed heavily on household confidence (**Figure 5**) and UK growth was either weak or negative through most of 2022-23 (**Figure 7**).

Figure 5

GfK Consumer Confidence index, 2020-2023

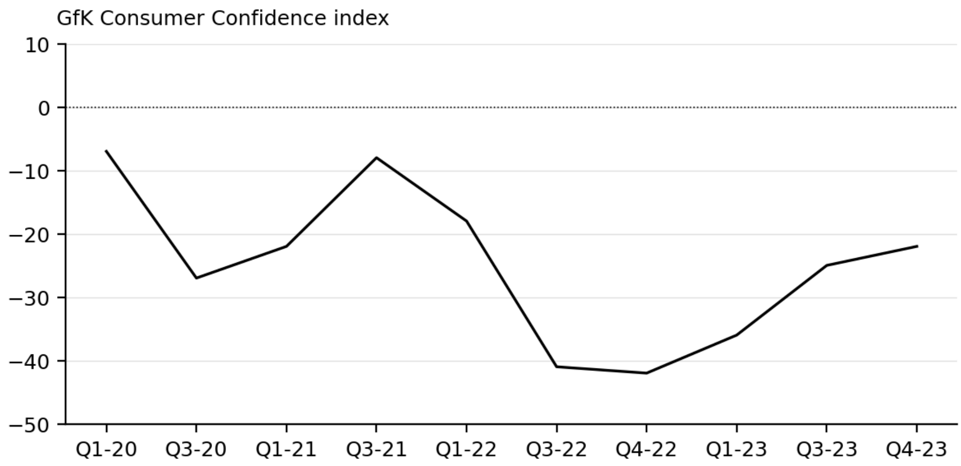


Figure 6

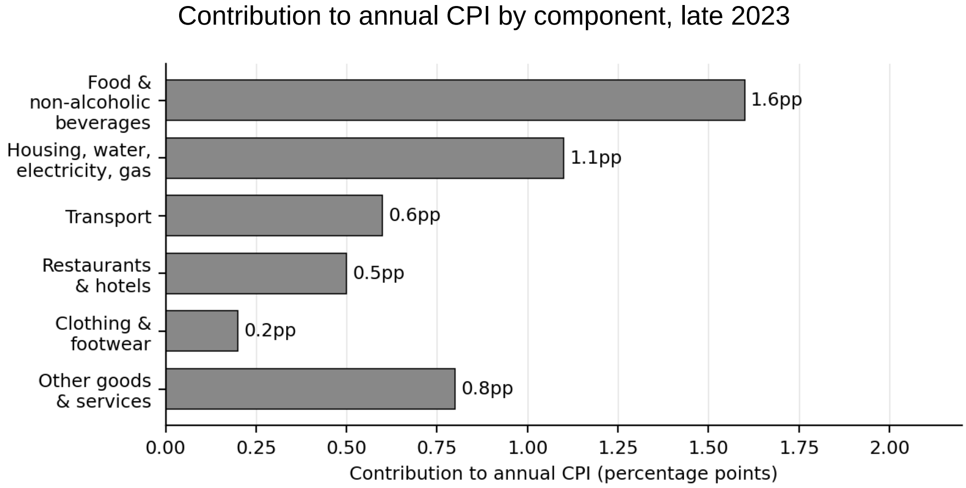
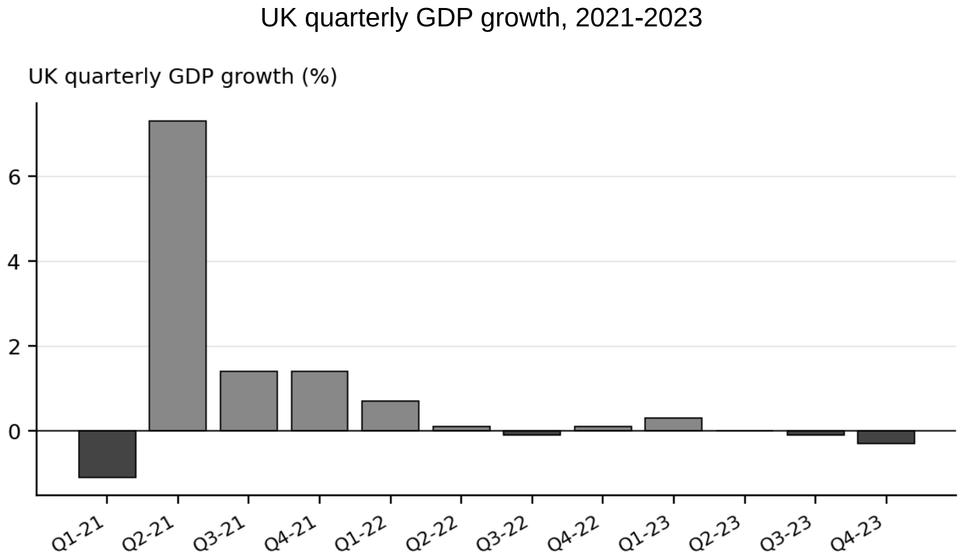


Figure 7



The fiscal-monetary mix

The Bank of England's tightening was partly offset by fiscal measures designed to cushion households from rising costs. A summary of the main elements of the policy response is shown in **Figure 8**.

UK macroeconomic policy response, 2022-2024	
Monetary policy	Bank Rate raised from 0.1% (Dec 2021) to 5.25% (Aug 2023) – the steepest tightening cycle since 1989. Quantitative Tightening: the Bank of England began actively selling gilts from the asset purchase programme in late 2022.
Fiscal support	Energy Price Guarantee (Oct 2022 – June 2023): typical bills capped at £2,500/year at a cost of around £40 billion. Cost-of-living payments totalling £900 to households on means-tested benefits.

Supply-side	Increased childcare support (15 hours free childcare for 9-month-olds from April 2024) aimed at reducing economic inactivity. Reforms to Restart and Returnerships schemes to encourage over-50s back into the labour market.
Wage policy	National Living Wage rose 9.7% in April 2023 (to £10.42) and 9.8% in April 2024 (to £11.44). Public sector pay awards of 5-7% for most occupations during 2023-24.

Sources: ONS, Bank of England, HM Treasury, GfK.

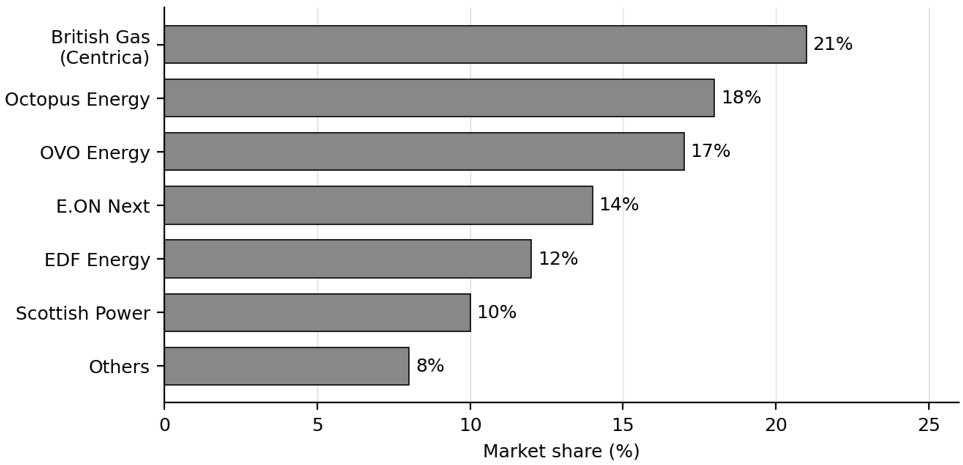
1 1	Using the data, explain the main causes of the rise in UK inflation between 2021 and 2022 (**Figure 1**).	[5]
1 2	Using the data, explain why the Bank of England raised Bank Rate so aggressively between December 2021 and August 2023 (**Figure 2**).	[5]
1 3	How effective is the UK's policy response (**Figure 8**) likely to be in returning inflation to the 2% target without causing a recession?	[10]
1 4	Using the data, discuss the extent to which the rise in Bank Rate is likely to reduce UK inflation (**Figures 1 and 2**).	[10]
1 5	Using the data, discuss whether the fall in real wages during 2022-23 (**Figure 3**) is likely to have a lasting effect on the UK economy.	[10]

The natural monopoly question

The UK domestic energy market was transformed during the 2010s and 2020s. Liberalisation in the 1990s had ended the regional monopolies held by the privatised successors to British Gas and the old Electricity Boards; by 2018 more than 70 independent suppliers competed for residential customers. Then came the energy price shock of 2021-22: wholesale gas prices rose nine-fold, the Ofgem price cap jumped sharply, and 29 suppliers failed in 2021 alone. The market has since consolidated around a smaller number of large firms (**Figure 1**), many of which display classic natural-monopoly characteristics.

Figure 1

GB domestic energy market shares, late 2023

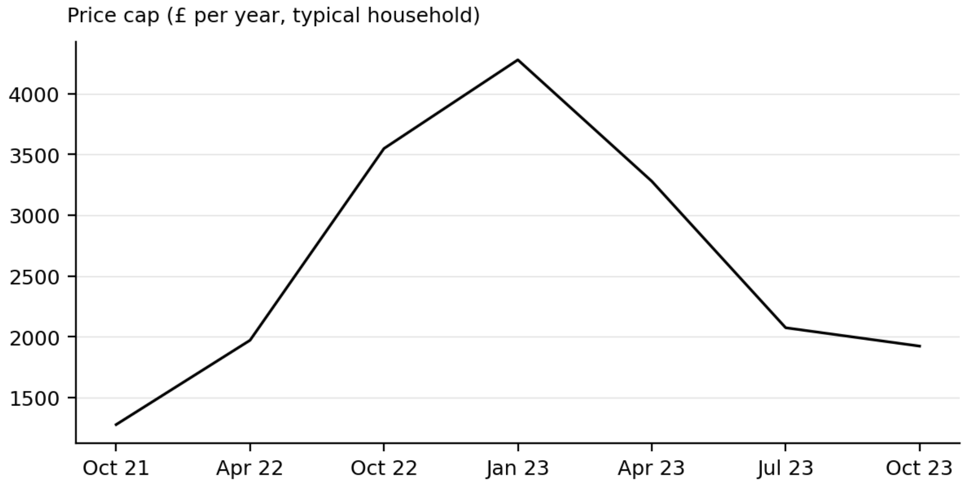


Some of the essential features of the market make it different from a typical competitive market. The electricity transmission and gas transmission networks are natural monopolies – it would be economically wasteful to duplicate the pylons, wires and pipes. Retail supply is nominally competitive, but customer acquisition is costly and the price cap introduced in 2019 sets an effective price ceiling for default-tariff customers. Economies of scale in procurement, hedging and operations strongly favour larger suppliers.

The Ofgem price cap is reviewed every three months and is based on a formula that passes through wholesale costs with a limited retailer margin. During the 2022 price shock the cap rose sharply, leaving suppliers squeezed between high wholesale costs and a cap that lagged behind (**Figure 2**).

Figure 2

Ofgem price cap, typical household (£/year), 2021-2023



The data below (**Figure 3**) shows the cost and revenue schedule for Power-Up Ltd, a medium-sized energy supplier. Demand is measured in thousands of megawatt-hours (MWh) sold per month.

Figure 3

Power-Up Ltd: Cost and revenue schedule

Price (£/ MWh)	MWh sold (000s)	Total Revenue	Marginal Revenue	Total Cost	Marginal Cost
£260	0			£2,000	
£240	10			£3,600	
£220	20			£5,000	
£200	30			£6,400	
£180	40			£7,800	
£160	50			£9,400	
£140	60			£11,200	
£120	70			£13,400	
£100	80			£16,200	

The market structure has important implications for consumers. First, the price cap protects most households from the worst of volatile wholesale prices – but it also limits the range of tariffs on offer, reducing the benefit of active switching. Second, the failure of 29 smaller suppliers in 2021 imposed large costs on remaining customers through the Supplier of Last Resort levy and the Energy Company Obligation, concentrating risk among the largest firms. Third, the natural-monopoly elements of the sector (networks, metering) are subject to price controls rather than market competition.

Is the current market structure in consumers' interests? Supporters argue that a stable market of large, well-capitalised suppliers is more resilient and able to pass through efficiency gains. Critics point to the limited scope for genuine price competition, the high concentration after consolidation, and the opacity of how wholesale costs are passed through to the cap. The CMA completed a market investigation in 2016 and Ofgem is regularly consulted on reform – including proposals for social tariffs, dynamic pricing and expanded heat-pump tariffs as Net Zero approaches.

Sources: Ofgem; Cornwall Insight; ONS.

2 1 Using the information provided on the UK energy market, outline the characteristics of a natural monopoly. [5]

2 2 Using **Figure 3**, calculate the marginal revenue (MR) and marginal cost (MC) for each of the output levels (MWh sold) from 0 to 80,000. Record the answers in your **pink** answer booklet. [7]

Using your answers explain at what level of output Power-Up Ltd will, in theory, maximise profits.

2 3 Using a costs and revenue diagram, evaluate the effects on an energy supplier's abnormal profits of Ofgem tightening the price cap. [10]

2 4 Evaluate the view that price regulation (such as the Ofgem price cap) is the best way to protect consumers in a natural-monopoly industry. [7]

2 5 With reference to the data, discuss the extent to which the consolidation of the UK energy market since 2021 reduces economic welfare. [11]

END OF PAPER